

Week 2: BM25 Retrieval Baseline

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1. Objective

Reproduce the standard MS MARCO Passage Ranking BM25 baseline using the bm25s pure-Python backend, against the official corpus and dev/small qrels.

2. Dataset

- **Corpus:** MS MARCO Passage Ranking collection (~8.8M passages)
- **Queries:** dev/small (6980 queries; 6980 have ≥ 1 positive qrel)
- **Relevance:** qrels.dev.small, binary judgments
- **Source:** loaded via `ir_datasets` (msmarco-passage and msmarco-passage/dev/small)

3. Method

- Tokenizer: bm25s default tokenizer with stopwords = en
- Parameters: $k1 = 1.50$, $b = 0.75$
- Top-1000 retrieved per query

Wall-clock:

- Indexing: — s
- Search: 4092.5 s (586.3 ms / query)

4. Evaluation Metrics

- MRR@10
- nDCG@10
- Recall@100
- Recall@1000

5. Results

Metric	Value
MRR@10	0.1703
nDCG@10	—
Recall@100	0.6212
Recall@1000	0.8154
# queries	6980

The published Anserini/Lucene BM25 baseline on this split is approximately **MRR@10 $\approx$ 0.184**. The number above should be in the same ballpark; small deviations are expected from tokenizer differences between Lucene and bm25s.

6. Case Studies

Sampled queries where BM25 places a relevant passage in the top-10:

203274 — hillsborough community college president

- First relevant passage rank: **5**
- Relevant doc id(s): 7953938

Top 3 retrieved:

1. [] (score=14.761) Hillsborough Community College. Hillsborough Community College (HCC) is a two-year community college, located in Hillsborough County, Florida.
2. [] (score=13.739) Hillsborough Community College president arrested on DUI charge. TAMPA, Fla. (WFLA) — The president of Hillsborough Community College faces a charge of drunk driving after a weekend arrest. Deputies say he was pulled ove...
3. [] (score=13.006) Hillsborough Community College President Kenneth Atwater, Hillsborough County Jail booking photo TAMPA, Fla. (WFLA) — The president of Hillsborough Community College faces a charge of drunk driving after a weekend arrest...

818842 — what is the difference in psychology and sociology

- First relevant passage rank: **2**
- Relevant doc id(s): 7907439, 7907440

Top 3 retrieved:

1. [] (score=13.257) Sociology vs Psychology. The difference between Sociology and Psychology is that sociology is the study of human behavior in groups while psychology is the study of the individual human mind. To be more elaborative, Soci...
2. [✓] (score=12.903) Sociology vs Psychology. The difference between Sociology and Psychology is that sociology is the study of human behavior in

groups while psychology is the study of the individual human mind. To be more elaborative, Soci...

3. [] (score=12.635) Sociology: In Sociology the focus is on the human society. Psychology: In Psychology, the focus is on the mental processes and behavior of the individual. Relationship: Sociology: Sociology establishes the relationship b...

1085327 — what county is ridgway pa in

- First relevant passage rank: 2
- Relevant doc id(s): 7119970

Top 3 retrieved:

1. [] (score=11.084) Find homes for sale and real estate in Ridgway, PA at realtor.com®. Search and filter Ridgway homes by price, beds, baths and property type.
2. [✓] (score=10.642) Ridgway is a borough in and the county seat of Elk County, Pennsylvania, United States. Ridgway was founded by Philadelphian shipping merchant Jacob Ridgway and James Gillis. Jacob Ridgway earned substantial wealth both...
3. [] (score=10.569) Copyright © 2017 The Ridgway Record | 325 Main Street, Suite A | Ridgway, PA 15853 | 814-773-3161. All property rights for the entire contents of this publication shall be the property of The Ridgway Record. No part here...

998381 — where is the willamette stone located?

- First relevant passage rank: 3
- Relevant doc id(s): 7826254

Top 3 retrieved:

1. [] (score=11.552) Willamette Stone and Willamette Meridian. Willamette Stone survey point. Survey marker for the Willamette Meridian at Willamette Stone State Park, Washington and Multnomah Counties. Land surveys accomplished under the U....
2. [] (score=10.139) Willamette Stone survey point. Survey marker for the Willamette Meridian at Willamette Stone State Park, Washington and Multnomah Counties. Oregon Parks and Recreation. Land surveys accomplished under the U.S. Government...
3. [✓] (score=9.992) The Willamette Meridian is one of 37 Principal Meridians in the United States, each of which is named, that define the rectangular survey system. The Willamette Stone is the Initial Point or Point of Beginning of the Wil...

288884 — how many miles in altitude is the united states satellite constellation

- First relevant passage rank: 1
- Relevant doc id(s): 7373639

Top 3 retrieved:

1. [✓] (score=14.530) A GPS satellite is a satellite used by the NAVSTAR Global Positioning System (GPS). The first satellite in the system, Navstar 1, was launched February 22, 1978. The GPS satellite constellation is operated by the 50th Sp...
2. [] (score=12.114) The GPS space segment consists of a constellation of satellites transmitting radio signals to users. The United States is committed to maintaining the availability of at least 24 operational GPS satellites, 95% of the ti...
3. [] (score=10.838) A geostationary satellite is an earth-orbiting satellite, placed at an altitude of approximately 35,800 kilometers (22,300 miles) directly over the equator, that revolves in the same direction the earth rotates (west to...

7. Error Analysis

Sampled queries where BM25 fails to surface any relevant passage in the top-10:

1093487 — laws synonyms

- Relevant doc id(s): 7743266

- BM25 top 3:

1. (score=6.702) Synonyms for constitution in Free Thesaurus. Antonyms for constitution. 50 synonyms for constitution: laws, code, charter, canon, body of law, state of health, build, body, make-up, frame, physique, physical condition, s...
2. (score=6.370) Princeton's WordNet(0.00 / 0 votes)Rate these synonyms: 1 government, governing, governance, government activity, administration(adj) the act of governing; exercising authority. Synonyms: 2 governing(adj) responsible f...
3. (score=6.284) Define legislative. legislative synonyms, legislative pronunciation, legislative translation, English dictionary definition of legislative. adj. 1. Of or relating to the enactment of laws. 2. Resulting from or decided by...

1016879 — what to use for foundation beams

- Relevant doc id(s): 7238544

- BM25 top 3:

1. (score=9.051) A pier and beam (sometimes called post and beam) foundation involves wood posts or concrete piers set into the ground and bearing the weight of the building on foundation beams. pier and beam (sometimes called post and b...
2. (score=8.861) Raft foundation. The extensive foundation (which is usually called raft foundation), is a unified foundation that extends throughout the entire area of the columns. As a rule, raft foundation is used as a building's foun...

3. (score=8.838) The extensive foundation (which is usually called raft foundation), is a unified foundation that extends throughout the entire area of the columns. As a rule, raft foundation is used as a building's foundation when the so...

1100224 — gorilla car washes in council bluffs iowa

- Relevant doc id(s): 7180062

- BM25 top 3:

1. (score=19.502) Gorilla Car Wash is a carwash located in Council Bluffs, Iowa. View phone number, website, employees, products, revenue, and more.
2. (score=19.216) Q: Where is Gorilla Car Wash located? A: Gorilla Car Wash is located at 3003 W Broadway, Council Bluffs, IA 51501. Q: How long has Gorilla Car Wash in Council Bluffs, IA been in business? A: Gorilla Car Wash was founded...
3. (score=17.864) Gorilla Car Wash is located at the address 3003 W Broadway in Council Bluffs, Iowa 51501. They can be contacted via phone at (712) 309-0069 for pricing, hours and directions. Gorilla Car Wash specializes in Hybrids, Wate...

1096656 — how many square feet does a student need?

- Relevant doc id(s): 7481700

- BM25 top 3:

1. (score=12.584) For 480 square feet of roof how many bundles of shingles do you need? How many square feet does a bundle of roofing shingles cover? About 32 square feet. An easy way to remember this is 1 bundle is needed for every 4' x...
2. (score=12.121) 1 67% - How much square footage does a pallet of bricks cover? 2 31% - How many 1 2x2 square feet ceramic tile will it take to cover 28 square feet? 3 38% - How many brick will it take to cover 912 square feet? 4 59% -...
3. (score=11.456) How many bundles of shingles for a 28000 square foot roof? One bundle of shingle covers about 33 square feet of roof hence around 850 bundles of shingles will be needed for 28000 square feet roof! How many square feet...

570070 — what are the four steps of the scientific method

- Relevant doc id(s): 7623736

- BM25 top 3:

1. (score=12.224) Steps of the Scientific Method. Learn the Steps of the Scientific Method. Experimentation is how you test a hypothesis in the scientific method. Experiments can be simple or complex. What matters is that you can control...

2. (score=11.964) Scientific Method Steps. As more proof that there is no one way to do science, different sources describe the steps of the scientific method in different ways. Some list three steps, some four and some five. Fundamentall...
3. (score=11.958) What are the eight steps of the scientific method? A: The eight steps of the scientific method are: identify the problem, ask questions, research the questions, create a hypothesis, conduct experiments, collec... Full An...

8. Limitations

- Pure lexical matching: BM25 cannot bridge surface vocabulary gaps. Failure cases concentrate on queries with paraphrasing, synonyms, or implicit reasoning.
- The dev/small qrels contain at most one positive judgment per query and are sparse, so true recall is likely underestimated.
- No hyperparameter tuning of k_1 / b .

9. Next Steps

- Add a Sentence-BERT bi-encoder retriever and a hybrid BM25 + dense fusion.
- Add a cross-encoder reranker over the BM25 top-100.
- Compare against the Anserini reference implementation to quantify the tokenizer gap.